TUNA Story

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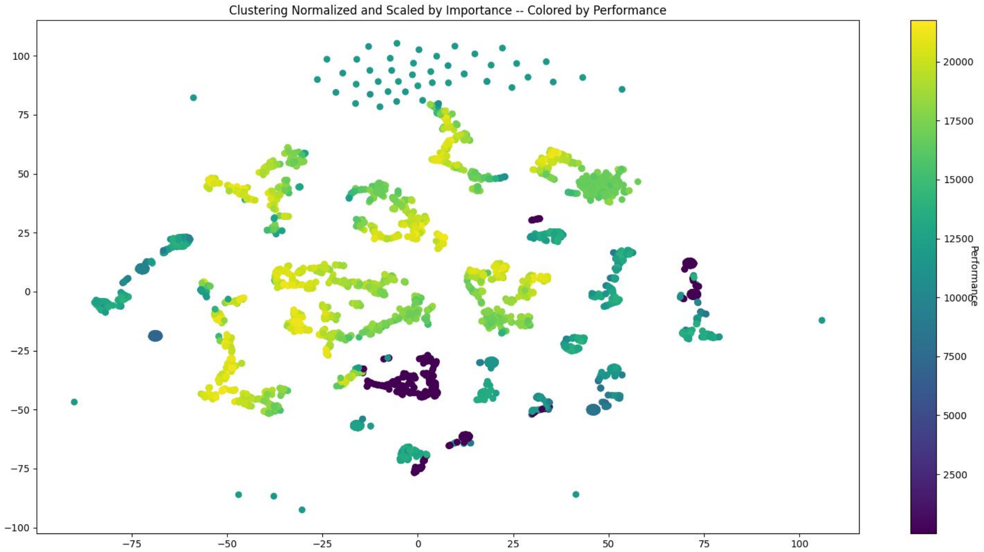
Original Research Questions

- Are all "good" configurations similar?
- Are there configurations which are more resilient to noise?

Are good Configurations Different?

Setup

- Take a series of configurations from tuning runs
- Use t-SNE or UMAP to map configuration space to 2-dimensional projection

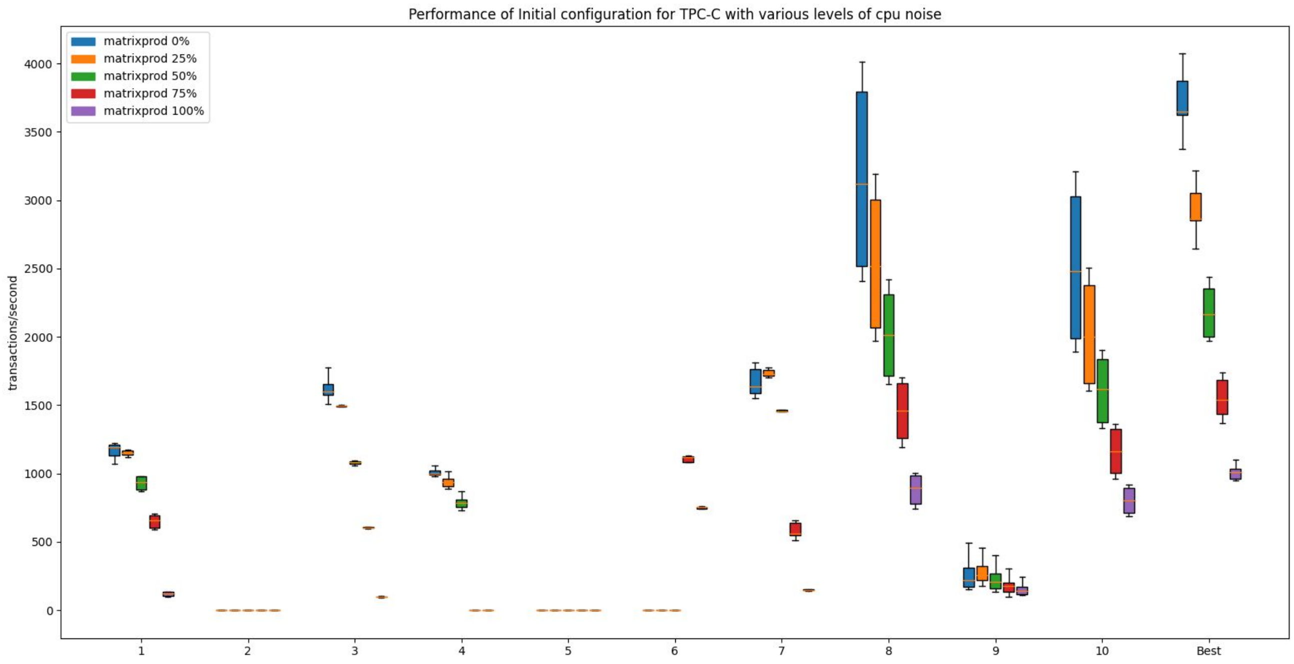


Are good Configurations Different?

TPC-C Cosine Distance between first 20 best configurations

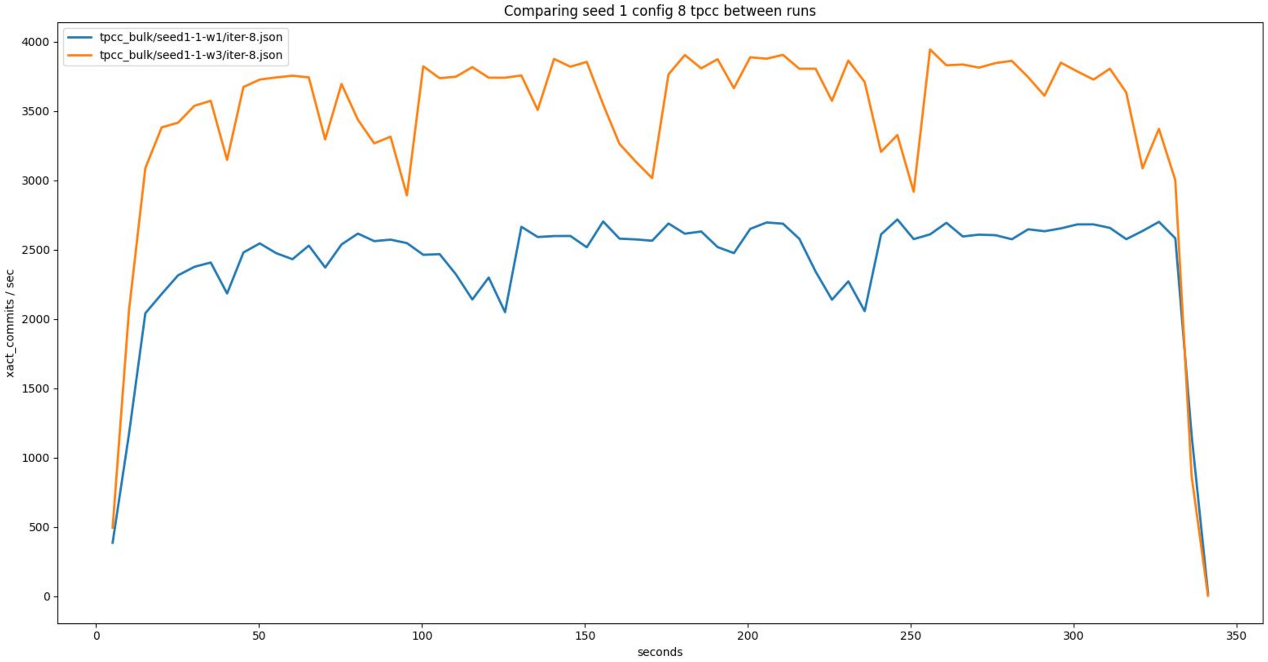
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	1.0	0.8477	0.2621	0.9785	0.7827	0.209	0.2213	0.8255	0.4635	0.2267	0.4644	0.2641	0.2146	0.2654	0.4655	0.8476	0.46	0.2774	0.2592	0.1741
2	0.8477	1.0	0.6386	0.8382	0.7209	0.2551	0.6452	0.9889	0.3909	0.5666	0.3912	0.6529	0.259	0.6509	0.3811	0.9971	0.3946	0.6453	0.6473	0.2293
3	0.2621	0.6386	1.0	0.1572	0.2285	0.279	0.9605	0.6342	0.5518	0.8281	0.5559	0.9778	0.2847	0.9785	0.5557	0.6291	0.5582	0.9791	0.976	0.2387
4	0.9785	0.8382	0.1572	1.0	0.8003	0.1862	0.1524	0.8216	0.3071	0.1651	0.3061	0.1754	0.1899	0.1746	0.2981	0.8383	0.3048	0.1768	0.1691	0.1654
5	0.7827	0.7209	0.2285	0.8003	1.0	0.7209	0.2141	0.753	0.4139	0.5409	0.4096	0.2538	0.7213	0.234	0.3448	0.6794	0.4633	0.201	0.2038	0.7144
6	0.209	0.2551	0.279	0.1862	0.7209	1.0	0.2714	0.301	0.5165	0.734	0.5097	0.3197	0.9996	0.2913	0.4163	0.1872	0.587	0.2412	0.2492	0.9964
7	0.2213	0.6452	0.9605	0.1524	0.2141	0.2714	1.0	0.6198	0.4821	0.8021	0.481	0.9924	0.2774	0.9922	0.4729	0.6379	0.4792	0.9831	0.9925	0.2379
8	0.8255	0.9889	0.6342	0.8216	0.753	0.301	0.6198	1.0	0.3398	0.6066	0.3414	0.6284	0.3033	0.6238	0.3241	0.9812	0.3558	0.6127	0.6167	0.2821
9	0.4635	0.3909	0.5518	0.3071	0.4139	0.5165	0.4821	0.3398	1.0	0.5192	0.9988	0.5701	0.5288	0.5706	0.9913	0.3662	0.9935	0.5897	0.5551	0.4447
10	0.2267	0.5666	0.8281	0.1651	0.5409	0.734	0.8021	0.6066	0.5192	1.0	0.5191	0.8282	0.7343	0.8099	0.4613	0.5215	0.5724	0.7745	0.7843	0.7174
11	0.4644	0.3912	0.5559	0.3061	0.4096	0.5097	0.481	0.3414	0.9988	0.5191	1.0	0.5705	0.5213	0.571	0.9928	0.3663	0.9941	0.5909	0.5554	0.4373
12	0.2641	0.6529	0.9778	0.1754	0.2538	0.3197	0.9924	0.6284	0.5701	0.8282	0.5705	1.0	0.326	0.9993	0.5632	0.6425	0.5708	0.9941	0.9967	0.2797
13	0.2146	0.259	0.2847	0.1899	0.7213	0.9996	0.2774	0.3033	0.5288	0.7343	0.5213	0.326	1.0	0.298	0.4285	0.1915	0.5977	0.2486	0.2562	0.9951
14	0.2654	0.6509	0.9785	0.1746	0.234	0.2913	0.9922	0.6238	0.5706	0.8099	0.571	0.9993	0.298	1.0	0.5674	0.6429	0.5682	0.9969	0.9989	0.2499
15	0.4655	0.3811	0.5557	0.2981	0.3448	0.4163	0.4729	0.3241	0.9913	0.4613	0.9928	0.5632	0.4285	0.5674	1.0	0.364	0.9795	0.5951	0.5557	0.3401
16	0.8476	0.9971	0.6291	0.8383	0.6794	0.1872	0.6379	0.9812	0.3662	0.5215	0.3663	0.6425	0.1915	0.6429	0.364	1.0	0.3643	0.641	0.6423	0.1607
17	0.46	0.3946	0.5582	0.3048	0.4633	0.587	0.4792	0.3558	0.9935	0.5724	0.9941	0.5708	0.5977	0.5682	0.9795	0.3643	1.0	0.5822	0.5484	0.5183
18	0.2774	0.6453	0.9791	0.1768	0.201	0.2412	0.9831	0.6127	0.5897	0.7745	0.5909	0.9941	0.2486	0.9969	0.5951	0.641	0.5822	1.0	0.9979	0.1957
19	0.2592	0.6473	0.976	0.1691	0.2038	0.2492	0.9925	0.6167	0.5551	0.7843	0.5554	0.9967	0.2562	0.9989	0.5557	0.6423	0.5484	0.9979	1.0	0.2074
20	0.1741	0.2293	0.2387	0.1654	0.7144	0.9964	0.2379	0.2821	0.4447	0.7174	0.4373	0.2797	0.9951	0.2499	0.3401	0.1607	0.5183	0.1957	0.2074	1.0

Are there Configurations Which are Resistant to Noise?



Are there Configurations Which are Resistant to Noise?

Looking at one configuration deployed to two machines, there are no obvious reasons why a configuration performs worse



Why do configurations perform worse?

Machine	Run 8 Performance	Sequential Read IOPS	Sequential Bandwidth (MiB/s)	Random Read IOPS	Random Rand Bandwidth (MiB/s)	Sequential Write IOPS	Sequential Write Bandwidth (MiB/s)	Random Write IOPS	Random Write Bandwidth (MiB/s)
1	2389.81	330	332	55100	215	199	201	36600	143
2	3529.634	363	365	55100	215	193	194	39100	153
3	3530.606	356	358	55100	215	203	205	39400	154
4	3532.978	349	351	55100	215	205	207	39900	156
5	3535.803	349	351	54900	214	202	204	38600	151
6	2504.798	361	363	54100	211	178	180	39200	153
7	2380.102	356	358	54900	214	212	213	40100	157
8	3532.69	354	356	54100	211	215	217	40400	158
9	3541.18	357	359	54800	214	182	184	39400	154
10	3526.326	360	362	53500	209	205	207	39900	156

Why do configurations perform worse?

- One Query seems to be the main offender
- ```
  SELECT COUNT(DISTINCT (S_I_ID)) AS STOCK_COUNT
     FROM  order_line, stock
     WHERE OL_W_ID = $1
     AND OL_D_ID = $2
     AND OL_O_ID < $3
     AND OL_O_ID >= $4
     AND S_W_ID = $5
     AND S_I_ID = OL_I_ID
     AND S_QUANTITY < $6
  ```
- This query has three different plans that get generated and used at different frequencies depending on the machine
  ```

"Human Readable" Plans

Plan 2 (bad)

```
aggdistinct: SORTGROUPCLAUSE
lefttree:
      NESTLOOP
   lefttree:
      INDEXSCAN
      indexqual
         CONST
      indexqualorig
         CONST
righttree:
      INDEXSCAN
   indexqual
      CONST
   indexqualorig
      CONST
joinqual
```

Plan 3 (good?)

```
aggdistinct: SORTGROUPCLAUSE
lefttree:
      NESTLOOP
   lefttree:
      INDEXSCAN
      indexqual
         CONST
      indexqualorig
         CONST
righttree:
      INDEXSCAN
   indexqual
      PARAM
   indexqualorig
      PARAM
allParam:
      nestParam:
            NESTLOOPPARAM
```

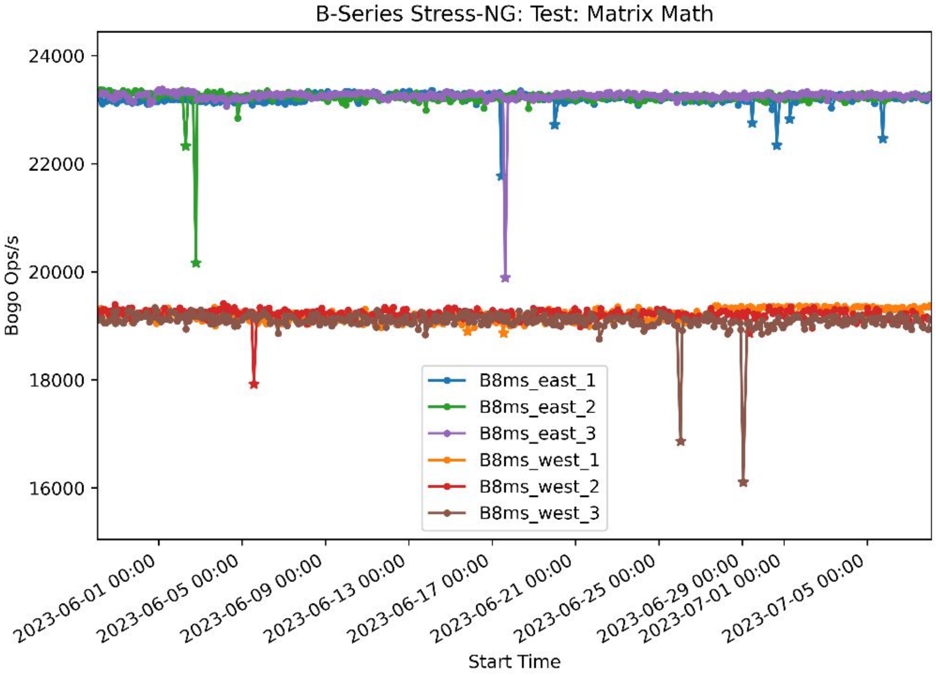
Plan 5 (good?)

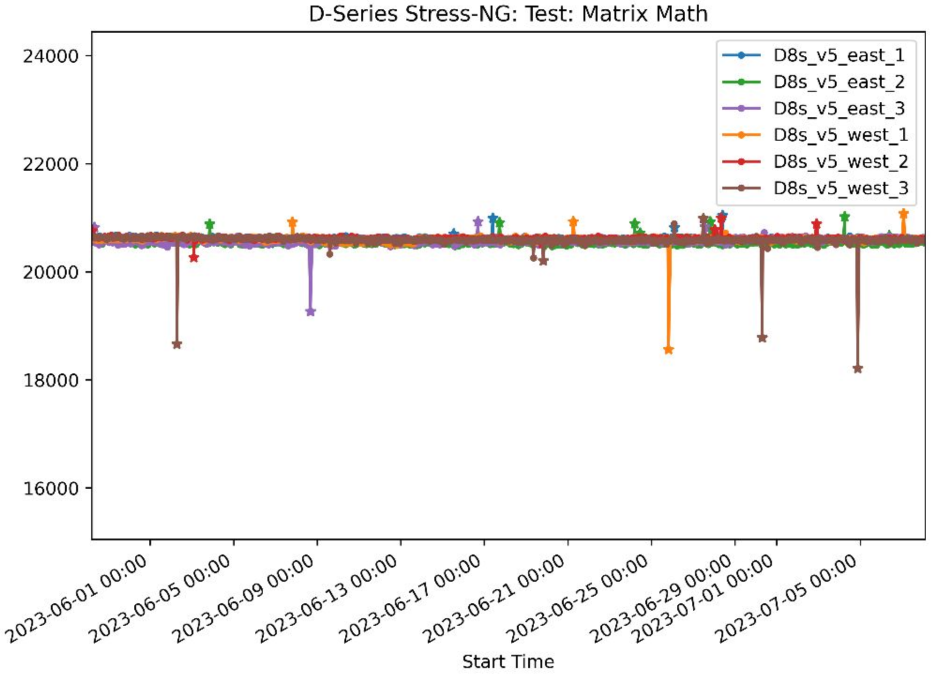
```
aggdistinct: SORTGROUPCLAUSE
lefttree:
      NESTLOOP
   lefttree:
      INDEXSCAN
righttree:
      INDEXSCAN
      indexqual:
         CONST
      indexqualorig:
         CONST
joinqual
```

Modified Research Questions

- Are all "good" configurations similar?
  - No
- ~~Are there configurations which are more resilient to noise?~~
- How much performance variability exists in cloud deployments?
  - How much does it matter?
- How do you prevent unstable configurations from small amounts of performance variability?

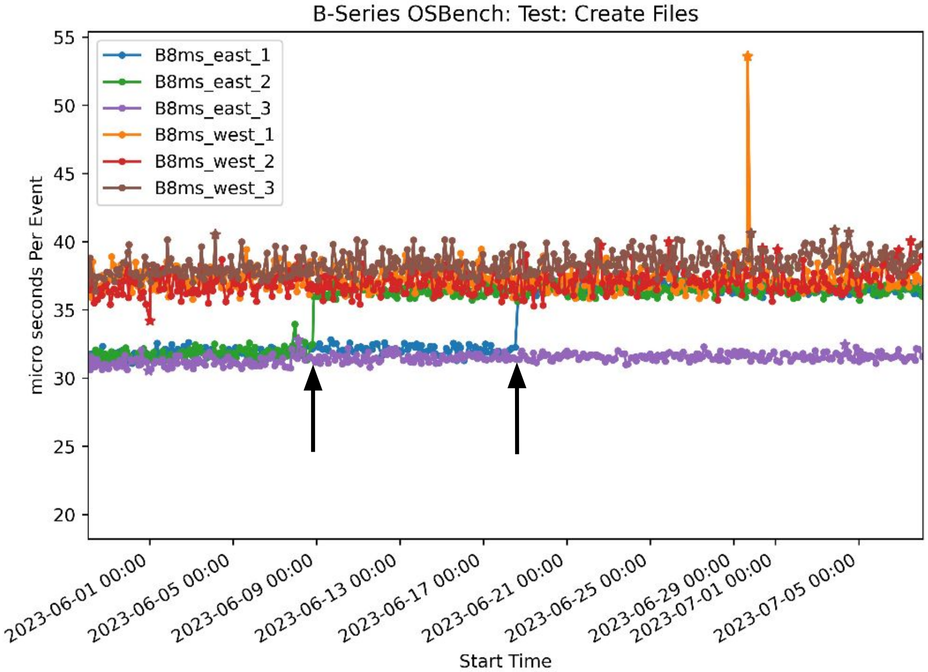
Regional Differences

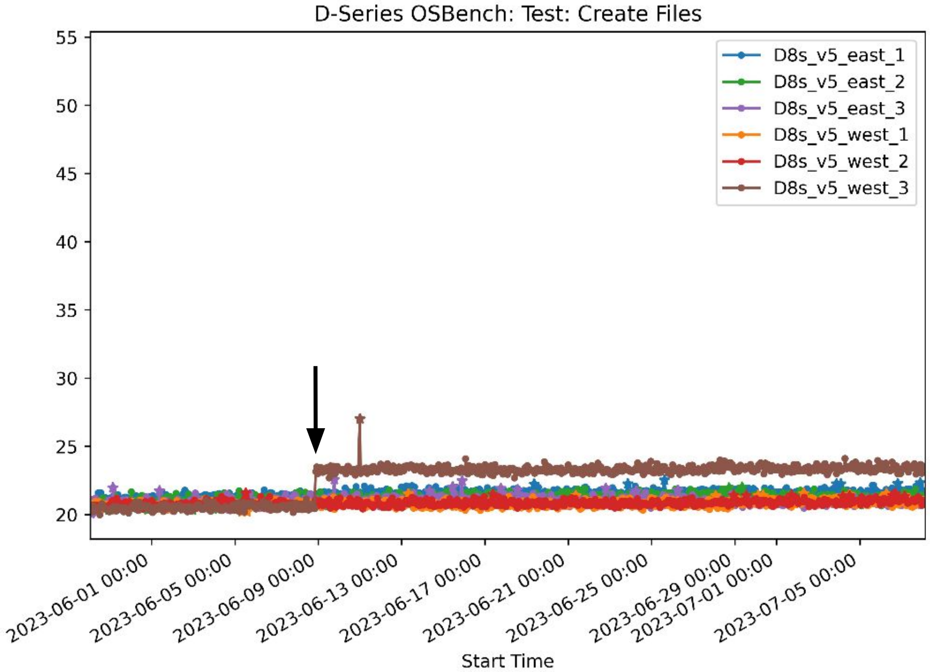




17.6% Performance differences between regions

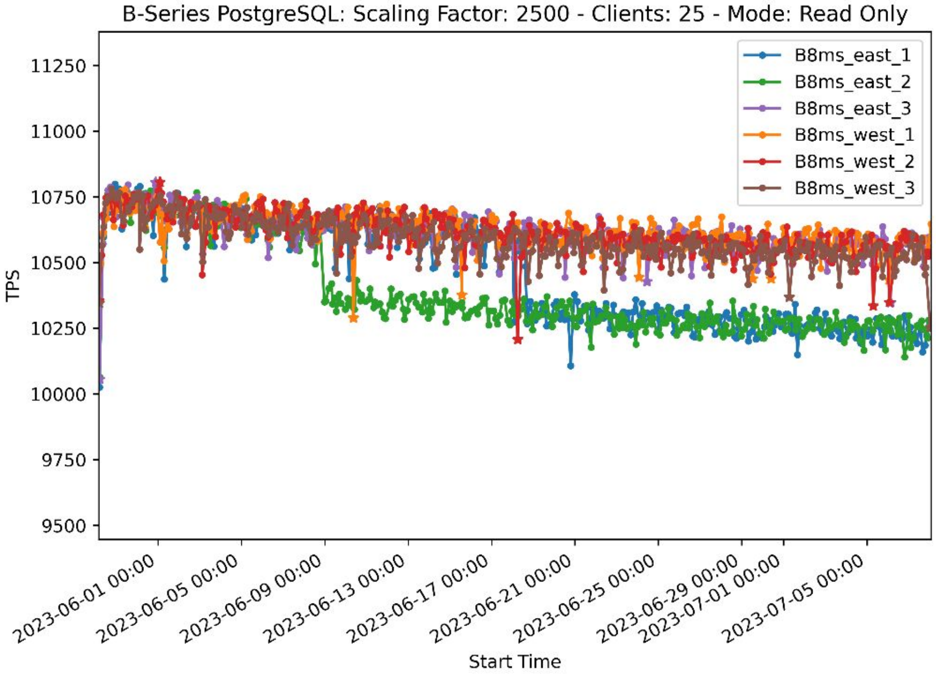
Sudden Performance Degradation

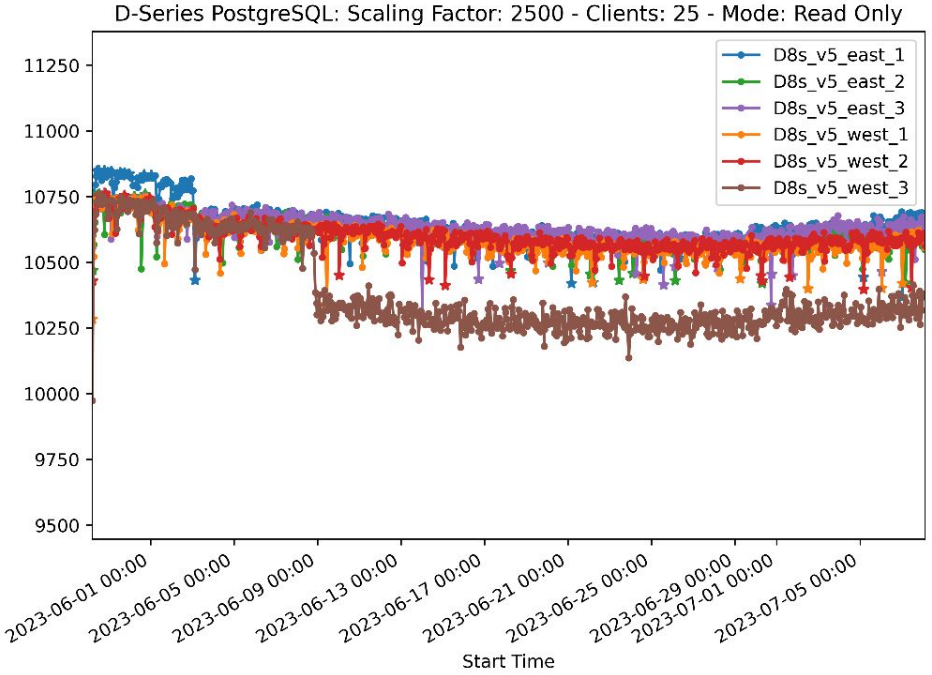




Sudden 12.5% Performance Degradation

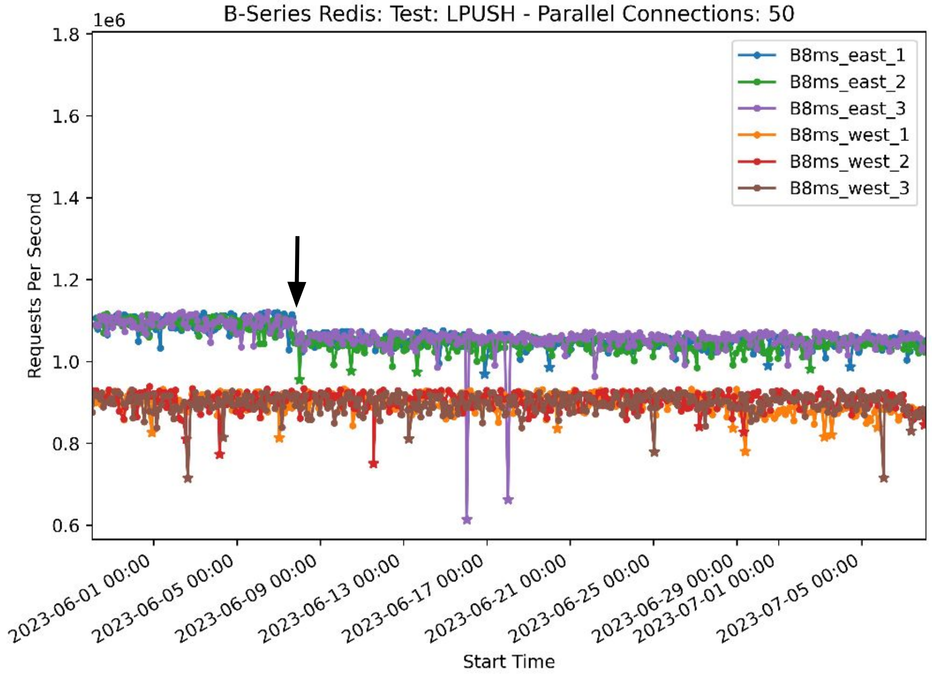
Slow Performance Degradation

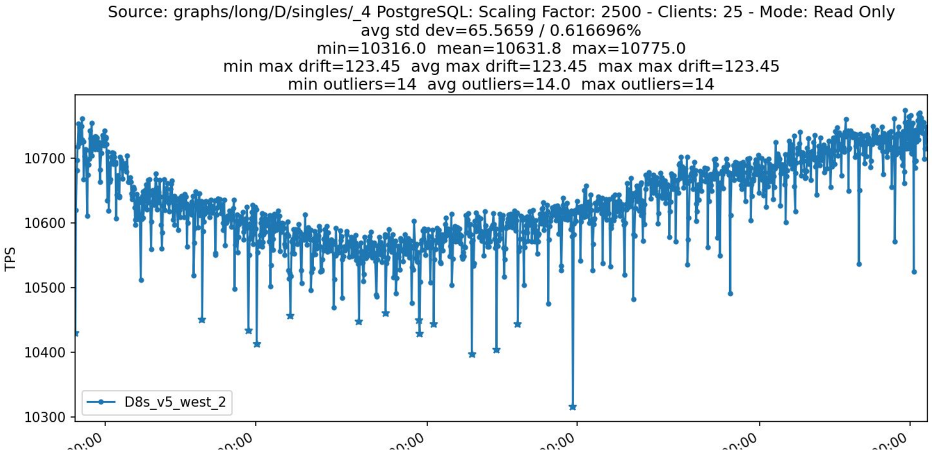


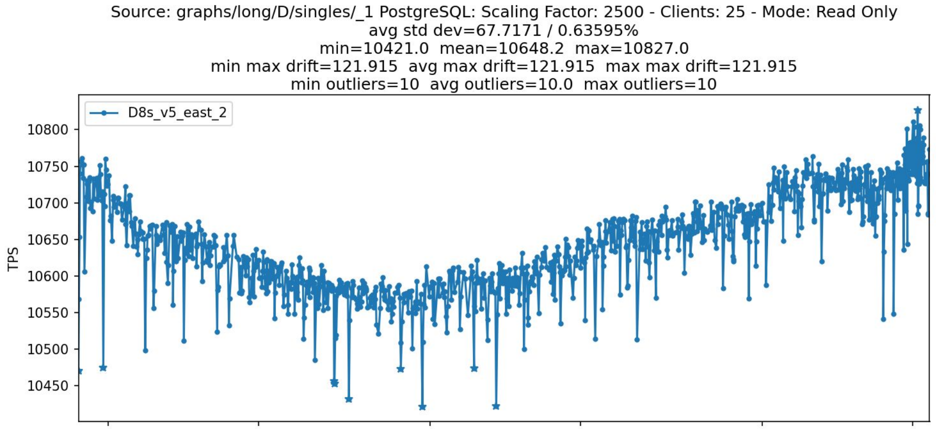


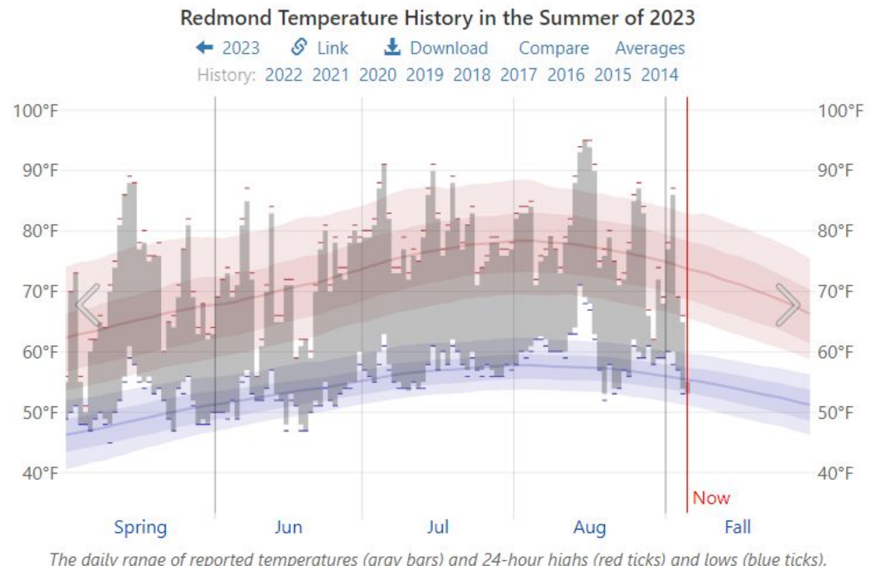
5% Performance Degradation over 35 days

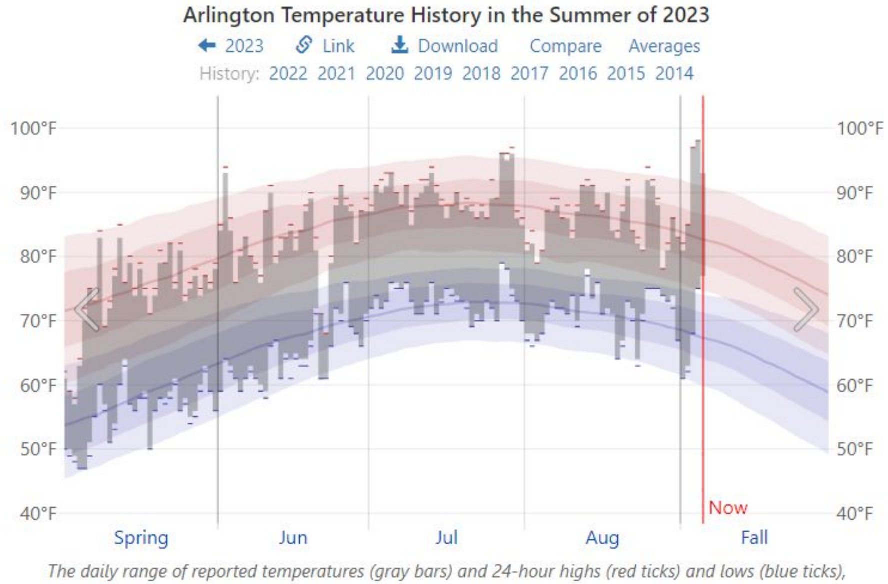
Sudden Performance Changes




Source: graphs/long/D/singles/_4 PostgreSQL: Scaling Factor: 2500 - Clients: 25 - Mode: Read Only
avg std dev=65.5659 / 0.616696%
min=10316.0 mean=10631.8 max=10775.0
min max drift=123.45 avg max drift=123.45 max max drift=123.45
min outliers=14 avg outliers=14.0 max outliers=14
TPS
10700
10600
10500
10400
10300
D8s_v5_west_2


Source: graphs/long/D/singles/_1 PostgreSQL: Scaling Factor: 2500 - Clients: 25 - Mode: Read Only
avg std dev=67.7171 / 0.63595%
min=10421.0 mean=10648.2 max=10827.0
min max drift=121.915 avg max drift=121.915 max max drift=121.915
min outliers=10 avg outliers=10.0 max outliers=10
TPS
D8s_v5_east_2
10800
10750
10700
10650
10600
10550
10500
10450


Redmond Temperature History in the Summer of 2023
2023 Link Download Compare Averages
History: 2022 2021 2020 2019 2018 2017 2016 2015 2014
100°F
90°F
80°F
70°F
60°F
50°F
40°F
Now
Spring
Jun
Jul
Aug
Fall
The daily range of reported temperatures (gray bars) and 24-hour highs (red ticks) and lows (blue ticks),


Arlington Temperature History in the Summer of 2023
2023 Link Download Compare Averages
History: 2022 2021 2020 2019 2018 2017 2016 2015 2014
100°F
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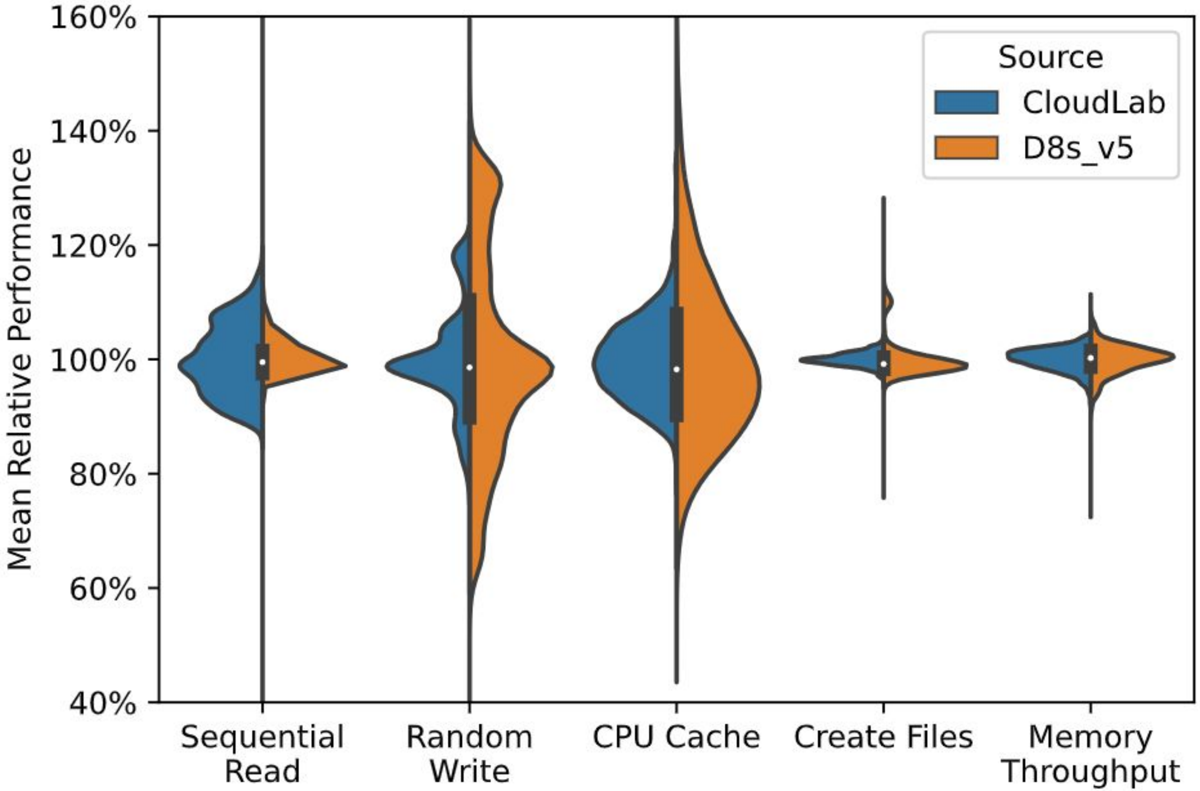
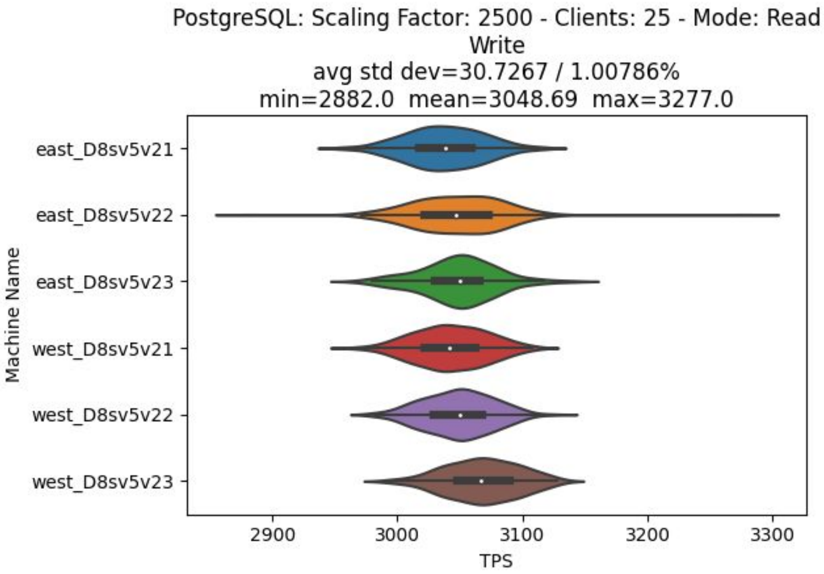
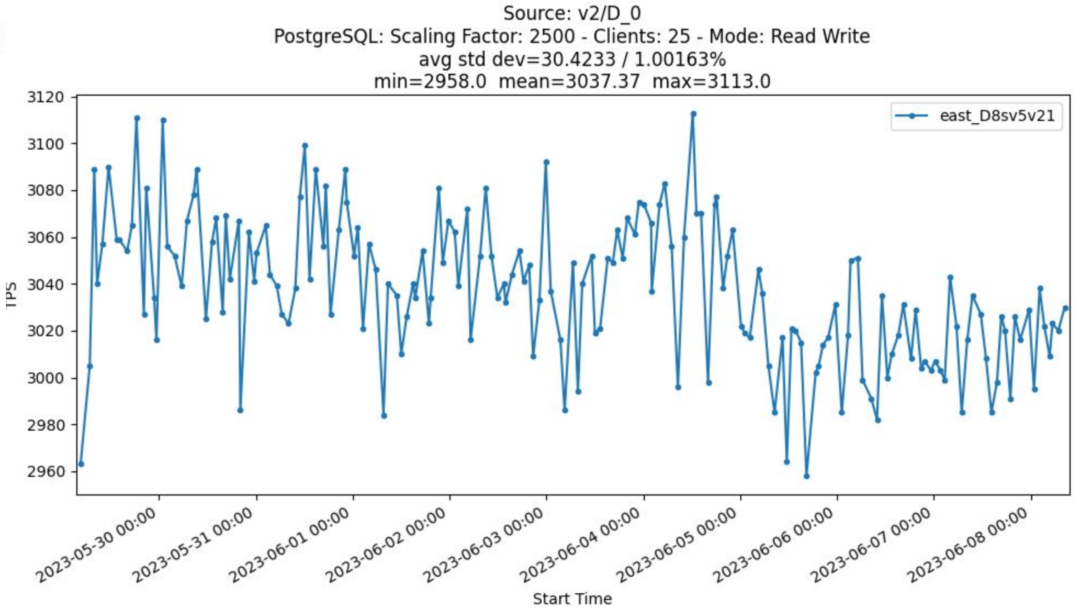
The daily range of reported temperatures (gray bars) and 24-hour highs (red ticks) and lows (blue ticks),

Bare Metal vs D series



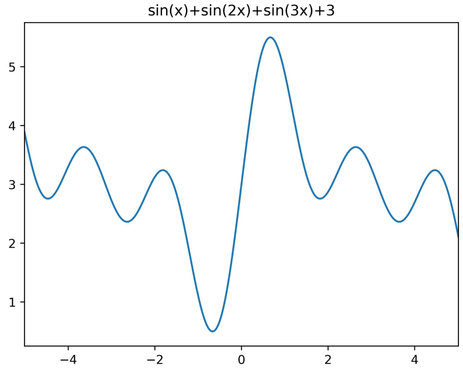
Performance Variability of End-to-End Apps

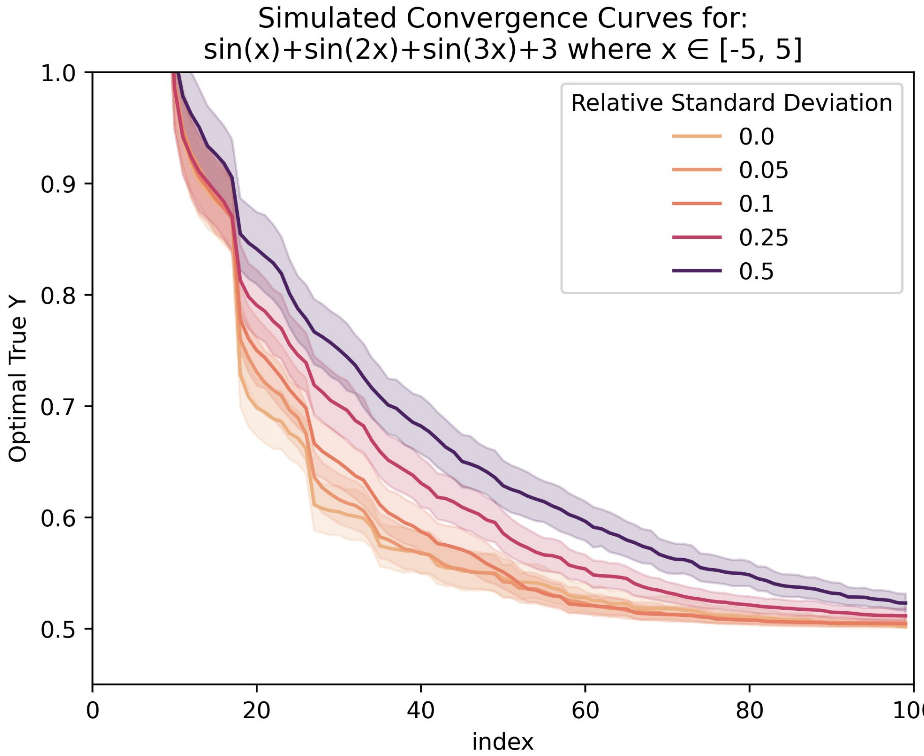




Convergence Simulation

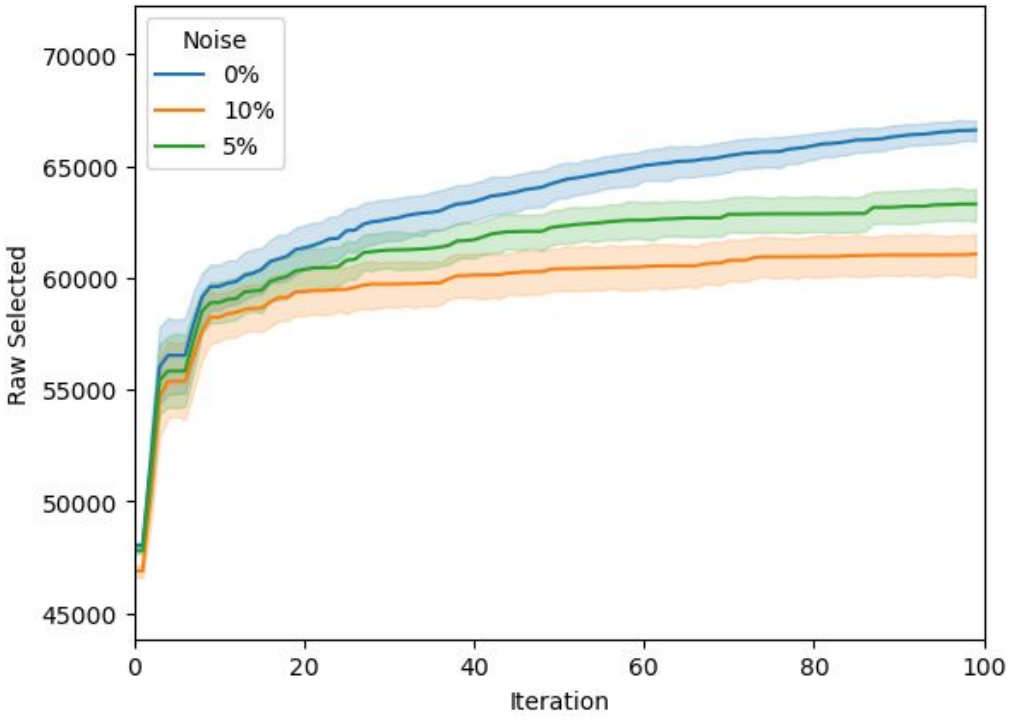
- 1000 samples per level of noise, each with a unique seed
- Implies worse suggestions
- Does not show very clearly with 100 samples per level of noise





Convergence of Cloudlab with Injected Gaussian Noise

- 20 Machines
- 5 Runs
- Single Machine Setup
- Postgres 16.1
- Injected with Gaussian Prior with 0%, 5%, and 10% noise
- Epinions workload selected for slow, but uniform convergence
- Ranges shown here are 95% confidence of mean (default of seaborn).



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  - Read the Paper